EPIDEMIQ

AI-Powered Epidemic Forecasting & Hotspot Detection

Team: NightRulers

01

PROBLEM STATAMENT:

Infectious diseases spread rapidly and unpredictably. Traditional monitoring systems are reactive — they tell us what has already happened, not what's coming. Data is fragmented across multiple repositories, making it hard to act quickly. The result? Delayed interventions, higher mortality, and massive social and economic disruption. What we need is a proactive, intelligent system that can unify data and provide early warnings

02

OUR SOLUTION

AN AI SYSTEM THAT

01 Predicts future cases

02 Detects hotspots

03 Provides risk levels

OBJECTIVES

1. Predictive Modeling: Forecast disease spread using time-series data.
2. Hotspot Identification: Detect high-risk regions early.
3. Risk Communication: Clear visual risk levels (Low/Medium/High).
4. Decision Support: Empower healthcare authorities, NGOs, and individuals.
5. Accessibility: Lower barriers for students & junior researchers.
6. Visuals: Icons for prediction, hotspot map, risk levels

HOW IT WORKS

Data Collection – Historical epidemic datasets

Data Preprocessing – Cleaning & structuring time-series data

Model Training – ML models for prediction

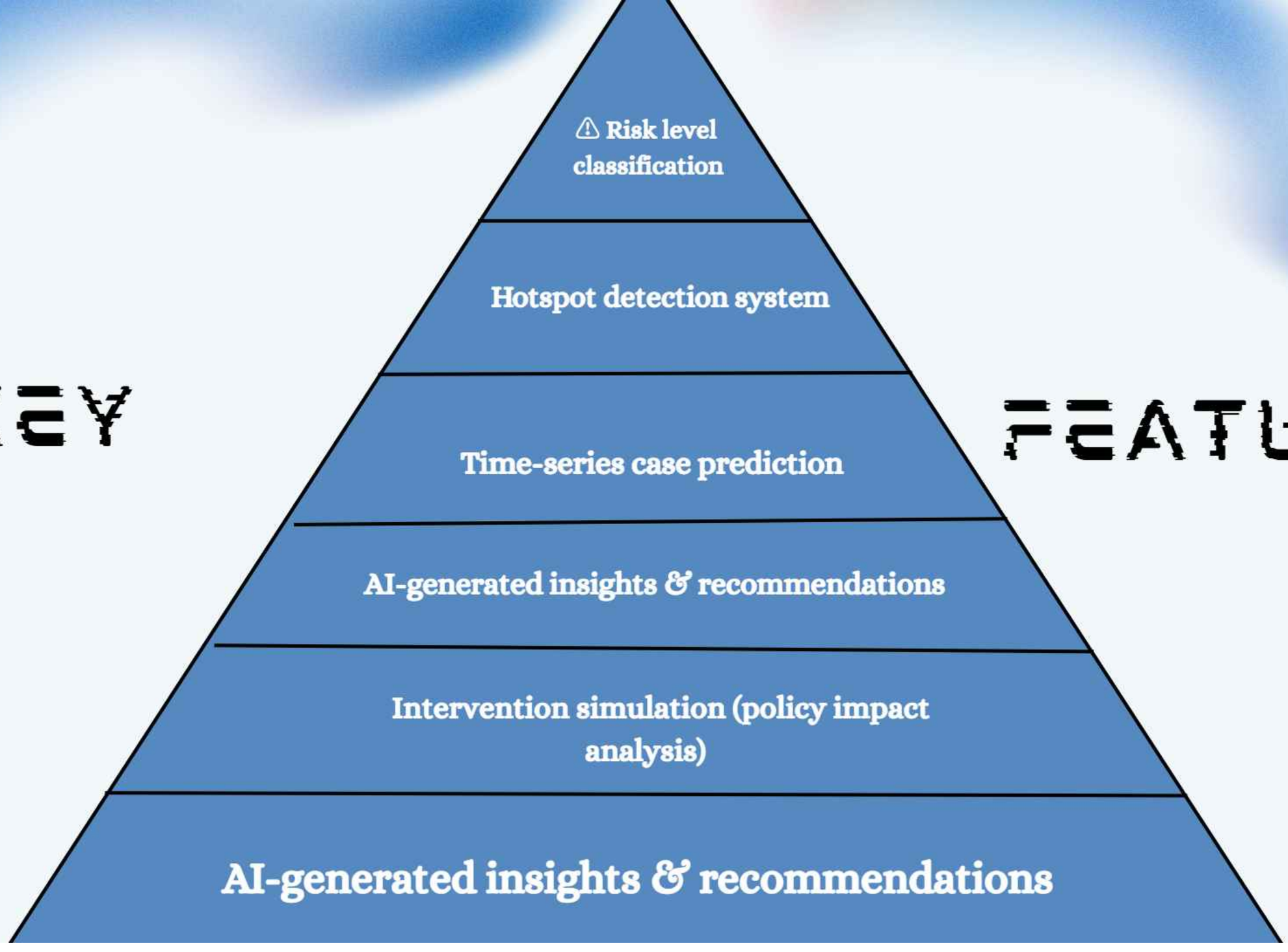
Forecasting – Generate future case predictions

Visualization – Display insights via dashboar

TECH STACK

- Frontend: React + TypeScript + Vite
- Styling: Tailwind CSS
- Animations: Motion (Framer Motion)
- Charts: Recharts
- AI Engine: Google Generative AI SDK (Gemini API)
- Backend: Firebase (Authentication & Firestore)
- Icons: Lucide React
- Data Parsing: PapaParse

KEY FEATURES



FUTURE ROADMAP

1. Collaborative Mode: Shared notebooks + peer review.
2. Granular Modeling: City-level + policy simulation.
3. Mobile App: Field researcher access.
4. Policy Simulation Engine: “What-if” scenarios for interventions.
5. Open-Source Expansion: Community-driven plugins.
6. Visuals: Roadmap timeline with milestones

IMPACT

- For Healthcare Authorities: Faster response, better resource allocation.
- For Researchers: Accessible modeling environment, reproducible results.
- For Students: Hands-on learning with real-world datasets.
- For Society: Early warnings → fewer lives lost, reduced economic disruption.
- Visuals: Infographic showing stakeholders + benefits

conclusion

EpidemIQ leverages AI to shift epidemic management from reactive to proactive, empowering authorities with tools for prediction, prevention, and preparednes s.

By NightRulers

Thank you!